Famicom JAMMA board

power

File: power.kicad_sch

Power: +5V entry, switched +5V_SW, +3V3, supply of the RP2A03 module

cpu

File: cpu.kicad_sch

CPU: RP2A03, work RAM, address decoder, data-bus resistors

ppu

File: ppu.kicad_sch

PPU: RP2C02, address latch, video RAM, sprite detection

cart

File: cart.kicad_sch

Cartridge connector and the dump circuit for emulation mode

video

File: video.kicad_sch

Video: RGB resistor DAC and composite sync

audio

File: audio.kicad_sch

Audio: mixer, loop through the cartridge, power amplifier

jamma

File: jamma.kicad_sch

JAMMA edge connector, inputs, coin counter and lock-out drivers

mcu

File: mcu.kicad_sch

Three Raspberry Pi Pico 2 modules, USB Type-C socket, links

PWR_FLAG

GND

PWR_FLAG

+5V

PWR_FLAG

+5V_SW

PWR_FLAG

VSYS

Sheet: /
File: fcjamma.kicad_sch

Title: Famicom JAMMA board

Size: A3	Date:	**Rev: 0.2**
KiCad E.D.A. 10.0.8		Id: 1/9

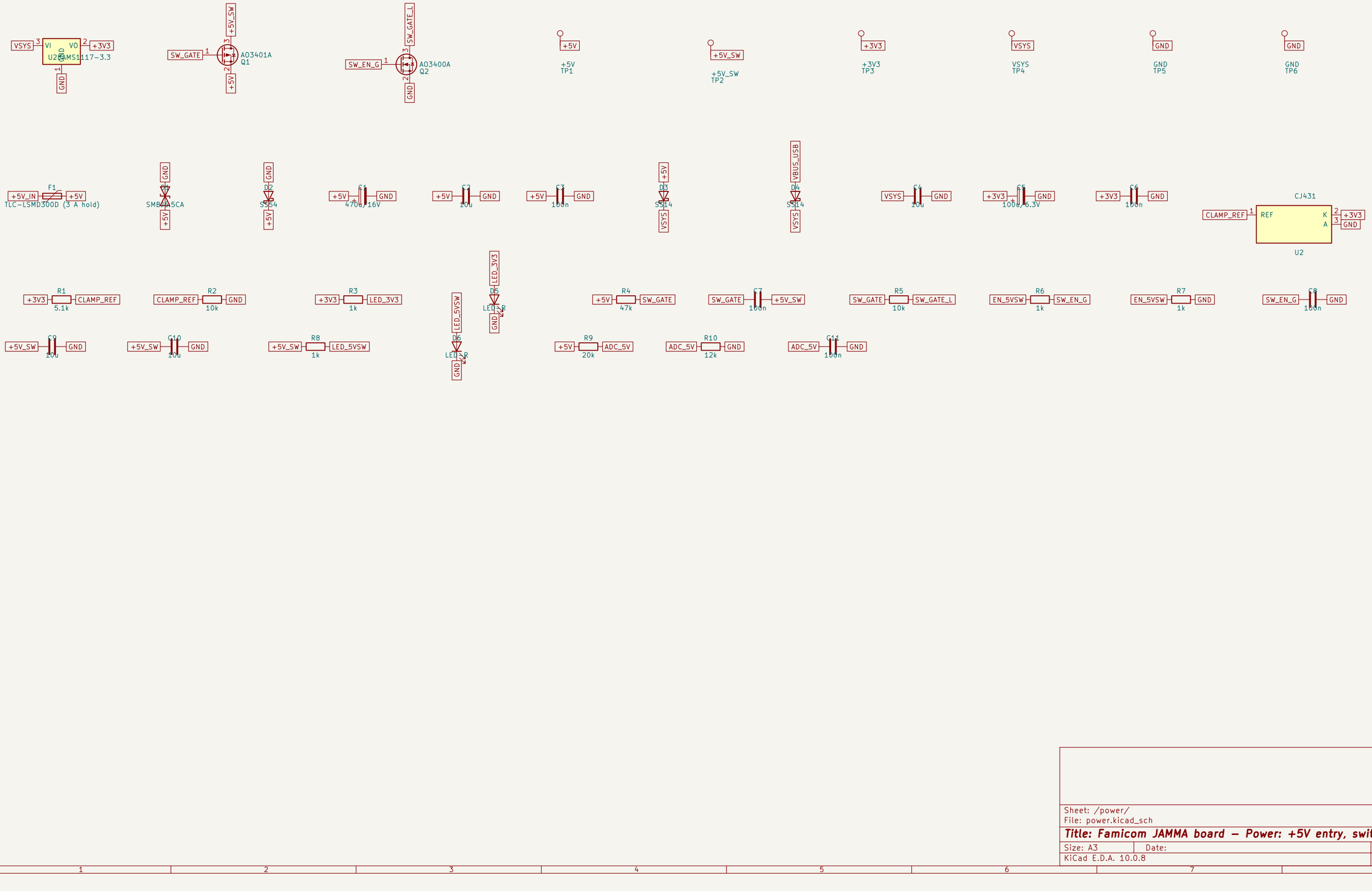
Power: +5V entry, switched +5V_SW, +3V3, supply of the Pico modules

+5V_IN comes from the JAMMA fingers 3, 4, C, D. −5 V is not used; +12 V only takes the turn-off current of the coils (sheet jamma).

+5V_SW feeds the CPU, the PPU, the memories, the 5 V logic and the cartridge. It is off until Pico C raises EN_5VSW,

so that no 5 V signal reaches a Pico pin before that Pico is powered (the RP2350 pins tolerate 5 V only while powered).

VSYS feeds the three Pico modules and the +3V3 regulator from +5V or from the USB socket, whichever is present.



Sheet: /power/
File: power.kicad_sch

Title: Famicom JAMMA board — Power: +5V entry, switched +5V_SW, +3V3, s

Size: A3	Date:	**Rev: 0.2**
KiCad E.D.A. 10.0.8		Id: 2/9

CPU: RP2A03, work RAM, address decoder, data-bus resistors

U1 is the donor's RP2A03, plugged into a 40-pin socket. In emulation mode the socket is empty and the dump shift registers drive the buses.

The 470 ohm resistors let Pico B replace what the PPU sees on a write without disturbing the CPU side of the bus.

No pull-ups on the CPU data bus: they are known to upset some PPU revisions, and games read the floating bus.

The logic next to the CPU and the PPU is 74HCT: its inputs switch at TTL levels, which is what the two NMOS chips deliver.

RP2A03 (donor)

U1

IS62C256AL (work RAM)

U3

74HCT138

U4

74HCT00

U5

CPU bus (option)

M2
TP7

CLK
TP8

RAM_A11-A14 (option)

J4

PPU: RP2C02, address latch, video RAM, sprite detector

U6 is the donor's RP2C02, plugged into a 40-pin socket. EXT0–3 give the lower four bits of the palette index of every pixel once

Pico B has forced bit 6 of register 2000 high. The fifth bit (sprite or background) is read from the brightness of VOUT:

Pico B makes the PPU store a dark colour in every background entry and a bright one in every sprite entry.

RP2C02 (donor)

74HCT573

U7

IS62C256AL (video RAM)

U8

PPU /RST = +5V (Famicom wiring)

JP1

PPU bus (option)

JP4

JP2

sprite bit source

MMBT3904

Q3

VBUF

TP9

VREF

TP10

CMP_OUT

TP11

composite video (test)

LMV7219

U9

Sheet: /ppu/	
File: ppu.kicad_sch	
Title: Famicom JAMMA board – PPU: RP2C02, address latch, video RAM, sprit	
Size: A2	Date:
KiCad E.D.A. 10.0.8	Rev: 0.2
	Id: 4/9

Cartridge connector and the dump circuit for emulation mode

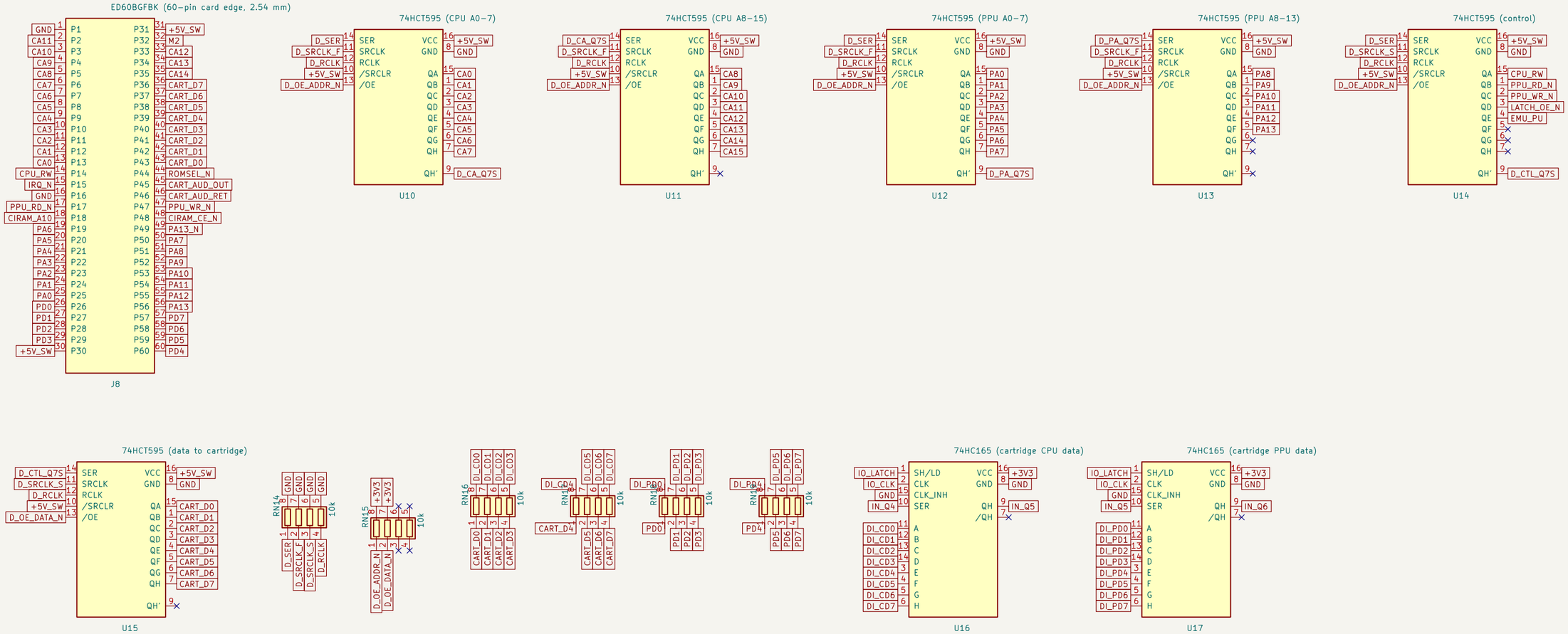
J8 is a new 60-pin card-edge connector. Pin 45 carries the board's sound to the cartridge, pin 46 brings it back.

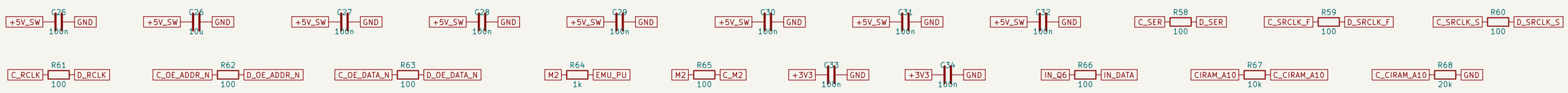
Dump circuit: with the CPU and PPU sockets empty, Pico C reads the cartridge through shift registers and hands the data to Pico A.

The six 74HCT595 run from +5V_SW and take the 3.3 V signals of Pico C directly (through 100 ohm).

In real-chip mode D_OE_ADDR_N and D_OE_DATA_N stay high (pull-ups) and every 74HCT595 output is switched off.

Pico C must hold its six lines to the 74HCT595 low or released whenever +5V_SW is off.





Sheet: /cart/
File: cart.kicad_sch

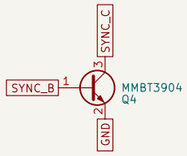
Title: Famicom JAMMA board – Cartridge connector and the dump circuit for

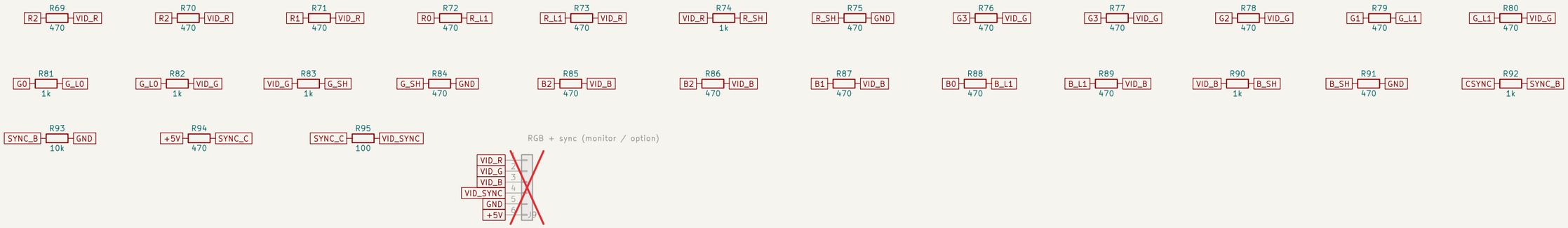
Size: A3	Date:	**Rev: 0.2**
KiCad E.D.A. 10.0.8		Id: 5/9

Video: RGB resistor DAC and composite sync

Pico A drives the DAC directly. Full scale is 3.0 V open circuit, about 2.85 V into 2 k, source resistance about 125 ohm.

Red and blue have 3 bits, green has 4. Only 470 ohm and 1 k are used so that the parts list stays short.





RGB + sync (monitor / option)

Sheet: /video/
File: video.kicad_sch

Title: Famicom JAMMA board – Video: RGB resistor DAC and composite sync

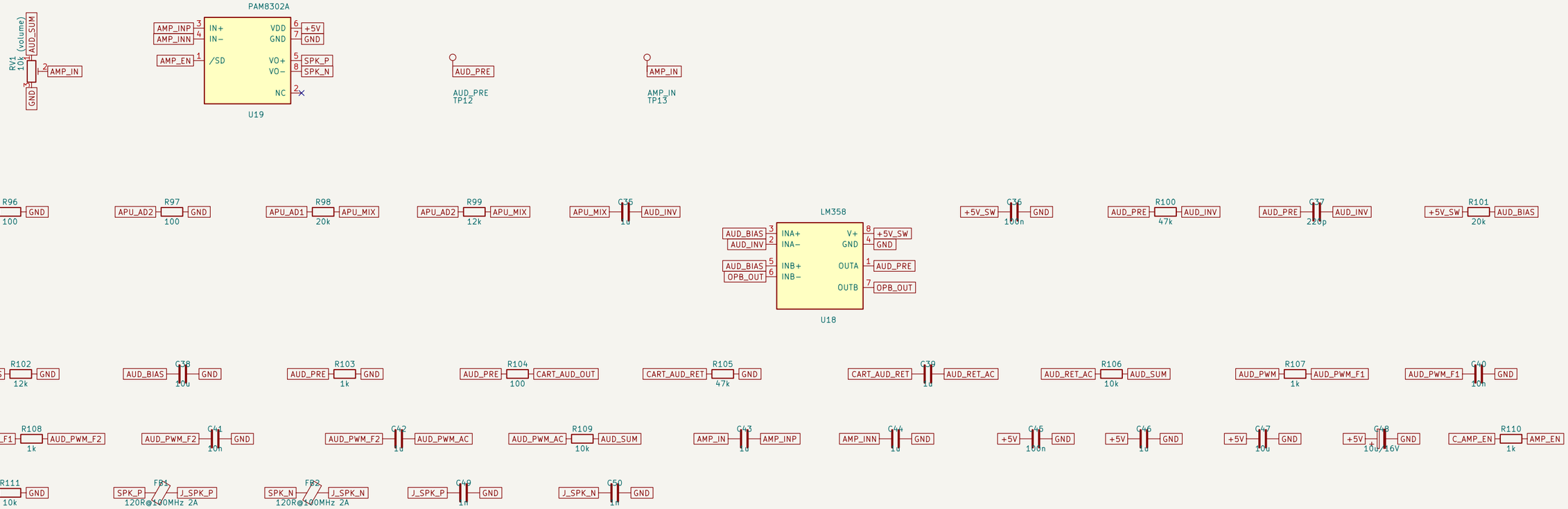
Size: A3	Date:	**Rev: 0.2**
KiCad E.D.A. 10.0.8		Id: 6/9

Audio: mixer, loop through the cartridge, power amplifier

Real-chip mode: 2A03 pins -> mixer (Famicom values) -> cartridge pin 45 -> cartridge -> pin 46 -> volume -> amplifier.

Emulation mode: Pico A's PWM output -> two RC stages -> volume -> amplifier.

The loudspeaker outputs are bridge-tied: JAMMA L is not ground. The amplifier stays off until Pico C raises AMP_EN.



Sheet: /audio/
File: audio.kicad_sch

Title: Famicom JAMMA board – Audio: mixer, loop through the cartridge, pow

Size: A3	Date:	**Rev: 0.2**
KiCad E.D.A. 10.0.8		Id: 7/9

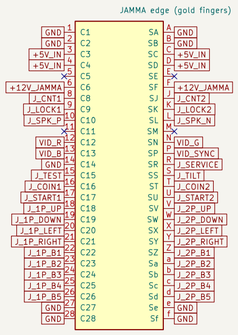
JAMMA edge connector, inputs, coin counter and lock-out drivers

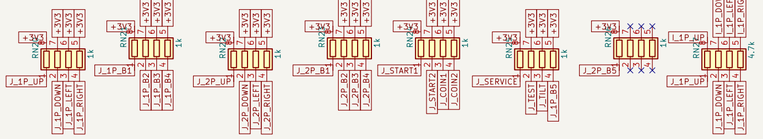
Every input: 1 k pull-up to +3V3 (3.3 mA through the switch), 4.7 k in series, 10 nF to ground, then a 74HC165 on +3V3.

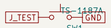
Pico C reads the chain: after a low pulse on IO_LATCH the first bit on IN_DATA is input H of U17, the last is input A of U20.

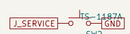
The same pulse copies the 74HC595's shift register to its outputs, so one cycle reads the inputs and writes the outputs.

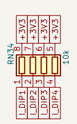
Fingers 5/E (−5 V) and 11/M are not connected. 6/F (+12 V) only take the turn-off current of the coils. 7/H is the key slot.

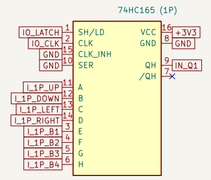


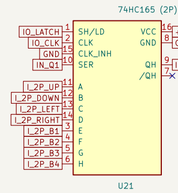


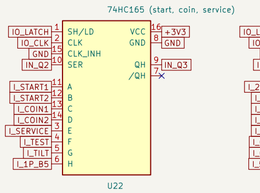


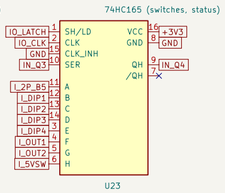


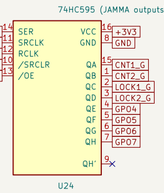


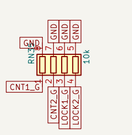


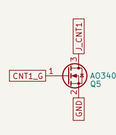






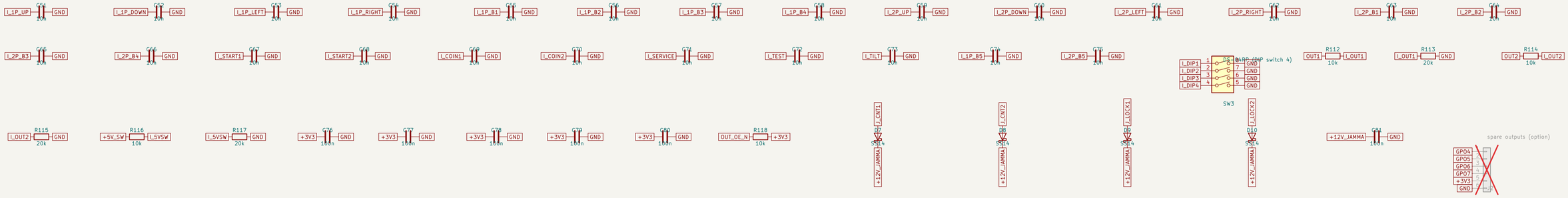












Sheet: /jamma/	
File: jamma.kicad_sch	
Title: Famicom JAMMA board — JAMMA edge connector, inputs, coin counter a	
Size: A2	Date:
KiCad E.D.A. 10.0.8	**Rev: 0.2**
	Id: 8/9

Three Raspberry Pi Pico 2 modules, USB Type-C socket, links

Pico A (video): clock and reset for CPU and PPU, palette index and sprite bit in, RGB and sync out; the emulator in emulation mode.

Pico B (bus): watches the CPU's accesses to the PPU, replaces palette data on the PPU side, answers the controller reads.

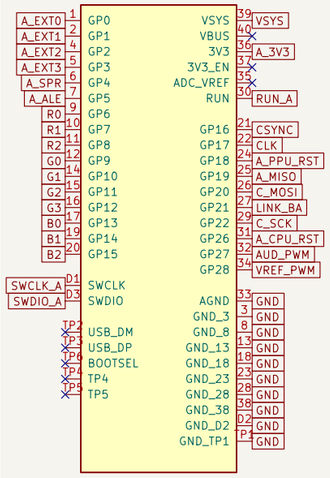
Pico C (input/output and management): JAMMA inputs and outputs, coin and timer logic, cartridge dump, USB Type-C,

power switch of the Famicom half, and reprogramming of Pico A and Pico B through their debug pads.

The modules are soldered flat on the board. The USB socket reaches Pico C through the pads under the module (TP2, TP3).

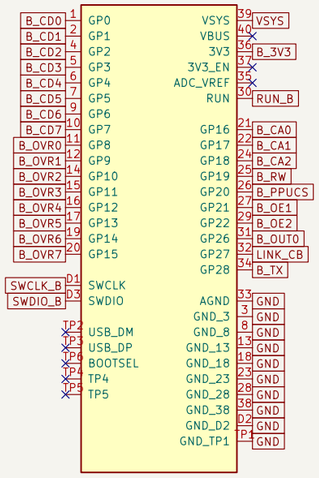
Use either the board's Type-C socket or Pico C's own Micro-USB socket, never both at once: they are the same two wires.

Pico 2 A (video)



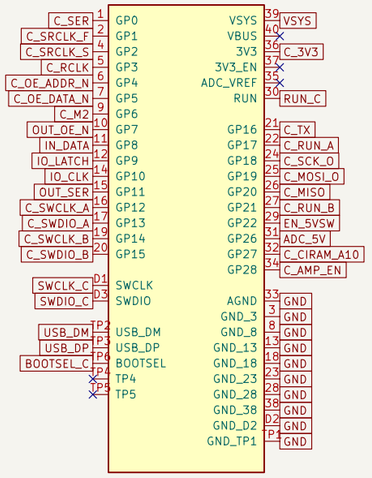
A_3V3
TP15

Pico 2 B (bus)



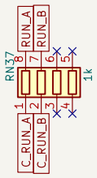
B_3V3
TP16

Pico 2 C (I/O, management)

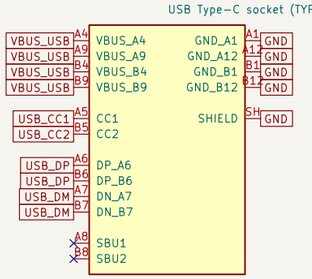


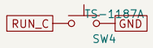
C_3V3
TP17

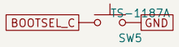




USB Type-C socket (TYPE-C16PIN)











USB lines (option)



Sheet: /mcu/
File: mcu.kicad_sch

Title: Famicom JAMMA board – Three Raspberry Pi Pico 2 modules, USB Type

Size: A3	Date:	**Rev: 0.2**
KiCad E.D.A. 10.0.8		Id: 9/9